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Inventor(s)

WU; Kai et al.

BATTERY CELL, BATTERY, AND ELECTRICAL DEVICE

Abstract

The battery cell includes a shell, the shell has a wall portion, the wall portion includes a weak portion, and the weak portion is configured to be destroyed when the battery cell releases internal pressure. An outer surface of the wall portion has a center point. A maximum distance between a projection of the weak portion on the outer surface of the wall portion and the center point is a , the minimum radial dimension of the outer surface of the wall portion is A , and a and A satisfy: $a \leq 0.4A$. By setting $a \leq 0.4A$, the weak portion is arranged at a region, with a relatively low stiffness, of the wall portion. When the battery cell releases the internal pressure, the region undergoes a large deformation under the action of gas, and therefore the weak portion arranged at the region is easily damaged.

Inventors: WU; Kai (Ningde, CN), CHEN; Xiaobo (Ningde, CN), LIU; Sike (Ningde, CN), GU; Mingguang (Ningde, CN), BAI; Jingfeng (Ningde, CN)

Applicant: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

Family ID: 93641389

Assignee: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International application PCT/CN2023/111081 filed on Aug. 3, 2023 that claims the priority to Chinese Patent Application No. 202310636914.6 filed on May 31, 2023. The content of these applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the field of batteries, and in particular, to a battery cell, a battery, and an electrical device.

BACKGROUND

[0003] Batteries are very widely used in the field of new energy, such as electric vehicles or new energy vehicles. New energy vehicles and electric vehicles have become a new development trend in the automobile industry. A battery is provided with a pressure relief mechanism for relieving internal pressure when the internal pressure of the battery reaches a fracture initiation pressure. However, the pressure relief mechanism is often opened prematurely, causing the normal pressure relief function to fail.

SUMMARY OF THE INVENTION

[0004] An objective of embodiments of the present application is to provide a battery cell, a battery, and an electrical device, which are intended to alleviate the problem in the related art of the pressure relief mechanism being often opened prematurely, causing the normal pressure relief function to fail.

[0005] In a first aspect, an embodiment of the present application provides a battery cell, where the battery cell includes a shell, the shell has a wall portion, the wall portion includes a weak portion, the weak portion is configured to be destroyed when the battery cell releases internal pressure, and an outer surface of the wall portion has a center point; where a maximum distance between a projection of the weak portion on the outer surface of the wall portion and the center point is a , the minimum radial dimension of the outer surface of the wall portion is A , a and A satisfy: $a \leq 0.4A$, and the radial direction refers to a direction passing through the center point.

[0006] In the above technical solution, by setting $a \leq 0.4A$, namely, the maximum distance between the projection of the weak portion on the outer surface of the wall portion and the center point of the outer surface of the wall portion is less than or equal to 0.4 times the minimum radial dimension of the outer surface of the wall portion, the weak portion is arranged in a first region, with a relatively small stiffness, of the wall portion. The first region is a circular region with the center point of the outer surface of the wall portion as a circle center and 0.4 times the minimum radial dimension of the outer surface of the wall portion as a radius. The first region is close to the center of the wall portion, and the first region is a region, with a relatively small stiffness, of the wall portion and thus has a relatively weak ability to resist deformation. When the battery cell releases the internal pressure, the first region undergoes a large deformation under the action of gas,

and therefore the weak portion arranged in the first region is easily deformed and damaged. Therefore, under the condition of the same fracture initiation pressure, the thickness of the weak portion arranged in the first region can be larger. In this way, when the battery cell is normally used, the weak portion has a stronger ability to resist external impact, which is helpful to reduce the risk of the weak portion being damaged prematurely. Furthermore, the thicker the weak portion is, the easier it is to manufacture, and the lower the precision requirement for a manufacturing device. [0007] As an optional technical solution of the embodiment of the present application, the wall portion is provided with a score groove, and the weak portion is formed at a bottom of the score groove.

[0008] In the above technical solution, the weak portion is formed by forming the score groove in the wall portion, with operation simple and convenient and costs low.

[0009] As an optional technical solution of the embodiment of the present application, the projection of the weak portion on the outer surface of the wall portion passes through the center point.

[0010] In the above technical solution, when the battery cell releases the internal pressure, the center point is a position, with the largest deformation, of the wall portion and is most likely to be damaged. By making the projection of the weak portion on the outer surface of the wall portion pass through the center point, the thickness of the weak portion at the center point can be larger. In this way, when the battery cell is normally used, the weak portion has a stronger ability to resist external impact, so as to reduce the risk of the weak portion being damaged prematurely.

[0011] As an optional technical solution of the embodiment of the present application, the weak portion has a weak point arranged corresponding to the center point, the thickness of the weak portion at the weak point is a first thickness, the maximum thickness of the weak portion at other positions than the weak point is a second thickness, and the first thickness is greater than or equal to the second thickness.

[0012] In the above technical solution, the first thickness is greater than the second thickness, namely, the thickness of the weak portion at the weak point is greater than the maximum thickness of the weak portion at other positions than the weak point. The thickness of the weak portion at the weak point is greater than the thickness of the weak portion at other positions, and the weak point has a stronger ability to resist external impact, so as to reduce the risk of the weak portion being damaged prematurely. The first thickness is equal to the second thickness, namely, the thickness of the weak portion at the weak point is equal to the maximum thickness of the weak portion at other positions than the weak point. In this way, the thickness of the weak portion is relatively more uniform, thereby facilitating processing and manufacturing.

[0013] As an optional technical solution of the embodiment of the present application, the score groove includes at least one groove section, the weak portion includes at least one weak section, and the at least one groove section and the at least one weak section are arranged in a one-to-one correspondence.

[0014] In the above technical solution, each time one groove section is arranged on the wall portion, one weak section is formed correspondingly. During manufacturing, quick forming can be achieved by stamping or milling, with operations simple and convenient and costs low.

[0015] As an optional technical solution of the embodiment of the present application, the at least one weak section includes at least one first weak section, the weak point is arranged at the first weak section, the thickness of the first weak section at the weak point is the first thickness, the thickness of the first weak section at other positions than the weak point is a third thickness, and the first thickness is greater than or equal to the third thickness.

[0016] In the above technical solution, the first thickness is greater than the third thickness, namely, the thickness of the first weak section at the weak point is greater than the thickness of the first weak section at other positions than the weak point. The thickness of the first weak section at the weak point is greater than the thickness of the first weak section at other positions, and therefore

the weak point has a stronger ability to resist external impact, so as to reduce the risk of the weak portion being damaged prematurely. The first thickness is equal to the third thickness, namely, the thickness of the first weak section at the weak point is equal to the thickness of the first weak section at other positions than the weak point. In this way, the thickness of the first weak section is relatively more uniform, thereby facilitating processing and manufacturing.

[0017] As an optional technical solution of the embodiment of the present application, the at least one weak section further includes a second weak section and a third weak section, the second weak section and the third weak section are arranged opposite to each other and spaced apart, the first weak section is configured to connect the second weak section and the third weak section, the thickness of the second weak section and the thickness of the third weak section are both a fourth thickness, and the fourth thickness is less than the third thickness.

[0018] In the above technical solution, when the battery cell releases the pressure, the weak portion can be opened along the first weak section, the second weak section, and the third weak section, forming a larger pressure relief area, which is beneficial to improve a pressure relief rate. The first weak section is closer to the center of the first region than the second weak section and the third weak section. The thickness of the first weak section at other positions than the weak point may also be greater than the thickness of the second weak section and the third weak section, thereby improving the ability of the weak portion to resist external impact and reducing the risk of the weak portion being damaged prematurely.

[0019] As an optional technical solution of the embodiment of the present application, one end of the first weak section is connected to a middle of the second weak section, the other end of the first weak section is connected to a middle of the third weak section, and the weak point is located at a midpoint of the first weak section.

[0020] In the above technical solution, when the battery cell releases the pressure, after being destroyed, the weak point can rupture along the first weak section toward the second weak section and the third weak section, respectively. After reaching a point of intersection of the first weak section and the second weak section, the weak point ruptures from the point of intersection in an extension direction of the second weak section toward both ends of the second weak section. After reaching a point of intersection of the first weak section and the third weak section, the weak point ruptures from the point of intersection in an extension direction of the third weak section toward both ends of the third weak section to achieve fast pressure relief.

[0021] As an optional technical solution of the embodiment of the present application, the second weak section includes a first end and a second end, the third weak section includes a third end and a fourth end, the first end and the third end are arranged opposite to each other, the second end and the fourth end are arranged opposite to each other, the first weak section is configured to connect the first end and the fourth end or connect the second end and the third end, and the weak point is located at the midpoint of the first weak section.

[0022] In the above technical solution, when the battery cell releases the pressure, after being destroyed, the weak point can rupture along the first weak section toward the second weak section and the third weak section, respectively. After reaching the point of intersection of the first weak section and the second weak section, the weak point ruptures from the point of intersection in the extension direction of the second weak section toward the other end of the second weak section. After reaching the point of intersection of the first weak section and the third weak section, the weak point ruptures from the point of intersection in the extension direction of the third weak section toward the other end of the third weak section so as to achieve fast pressure relief.

[0023] As an optional technical solution of the embodiment of the present application, the at least one weak section further includes a fourth weak section and a fifth weak section, the fourth weak section and the fifth weak section are arranged opposite to each other and spaced apart, the second weak section and the fourth weak section intersect at the first end, the third weak section and the fifth weak section intersect at the second end, the first weak section is configured to connect the

first end and the second end, and the weak point is located at the midpoint of the first weak section. [0024] In the above technical solution, when the battery cell release the pressure, after being destroyed, the weak point can rupture along the first weak section toward the first end and the second end, respectively. After reaching the first end, the weak point ruptures from the first end along the second weak section and the fourth weak section, respectively. After reaching the second end, the weak point ruptures from the second end along the third weak section and the fourth weak section, respectively, to achieve fast pressure relief.

[0025] As an optional technical solution of the embodiment of the present application, the at least one weak section further includes a sixth weak section, and the first weak section and the sixth weak section intersect at the weak point.

[0026] In the above technical solution, when the weak point serves as a point of intersection of the first weak section and the sixth weak section, the weak point has a more concentrated stress and thus is more likely to rupture. Therefore, the thickness of the weak portion at the weak point can be set larger, so that when the battery cell is normally used, the weak point has a stronger ability to resist external impact, so as to reduce the risk of the weak portion being damaged prematurely.

[0027] As an optional technical solution of the embodiment of the present application, the projection of the weak portion on the outer surface of the wall portion is a closed structure, and the closed structure surrounds the center point.

[0028] In the above technical solution, the weak portion is configured as the closed structure extending along a closed trajectory, so that when the internal pressure of the battery cell reaches the fracture initiation pressure, the internal pressure can push open the weak portion from all sides, forming a larger opening and allowing faster and easier pressure relief.

[0029] As an optional technical solution of the embodiment of the present application, in a thickness direction of the wall portion, the center of the projection of the weak portion coincides with the center point.

[0030] In the above technical solution, when the weak point coincides with the center point, the weak point has the largest deformation when the battery cell releases the pressure and thus is most likely to rupture. Therefore, the thickness of the weak portion at the weak point can be set larger so that when the battery cell is normally used, the weak point has a stronger ability to resist external impact, so as to reduce the risk of the weak portion being damaged prematurely.

[0031] As an optional technical solution of the embodiment of the present application, in the thickness direction of the wall portion, the center of the projection of the weak portion deviates from the center point.

[0032] In the above technical solution, by making the center of the projection of the weak portion in the thickness direction deviate from the center point, it is beneficial to improve the creep resistance of the weak portion.

[0033] As an optional technical solution of the embodiment of the present application, a distance at which the center of the projection of the weak portion in the thickness direction deviates from the center point is B, which satisfies: $0 < B \leq 5$ mm.

[0034] In the above technical solution, by limiting the distance at which the center of the projection of the weak portion in the thickness direction deviates from the center point in the range of 0-5 mm (excluding 0), the weak portion has better creep resistance.

[0035] As an optional technical solution of the embodiment of the present application, $0 < B \leq 3$ mm.

[0036] In the above technical solution, by limiting the distance at which the center of the projection of the weak portion in the thickness direction deviates from the center point in the range of 0-3 mm (excluding 0), the weak portion has better creep resistance. At the same time, the thickness of the weak portion is relatively large, so that when the battery cell is normally used, the weak portion has a stronger ability to resist external impact, which is beneficial to reduce the risk of the weak portion being damaged prematurely.

[0037] As an optional technical solution of the embodiment of the present application, the wall

portion includes a main body portion, and the weak portion and the main body portion are integrally formed.

[0038] In the above technical solution, integrally forming the weak portion with the main body portion can improve the stiffness of the weak portion, so that the weak portion has a stronger ability to resist creep and impact of an electrolytic solution, which is beneficial to prolong the service life of the weak portion and reduce the risk of liquid leakage occurring to the weak portion.

[0039] As an optional technical solution of the embodiment of the present application, the hardness of the weak portion is greater than that of the main body portion.

[0040] In the above technical solution, the weak portion can be formed by stamping, so that the weak portion will undergo cold work hardening (the grain arrangement changes, resulting in lattice distortion, reducing the metal plasticity, and increasing the material hardness). The hardness of the weak portion will be greater than that of the main body portion, so that the weak portion has an enhanced ability to resist external impact, and is not likely to be damaged by external impact. In addition, the weak portion has an enhanced ability to resist fatigue and is less likely to fail due to fatigue.

[0041] As an optional technical solution of the embodiment of the present application, the wall portion includes a main body portion and a pressure relief mechanism, the pressure relief mechanism is arranged separately from the main body portion and connected thereto, the weak portion is arranged on the pressure relief mechanism, and an outer surface of the main body portion is the outer surface of the wall portion.

[0042] In the above technical solution, the pressure relief mechanism is arranged separately from the main body portion and installed on the main body portion, so as to facilitate processing and manufacturing.

[0043] As an optional technical solution of the embodiment of the present application, the shell includes a case and an end cover, the case has an opening, and the end cover closes the opening and is fixed to the case; and the wall portion is the end cover or a wall of the case.

[0044] In the above technical solution, the wall portion may be the end cover, or a bottom wall of the case, or a side wall of the case, and the weak portion is arranged in the first region of the end cover, the bottom wall or the side wall. Under the condition of the same fracture initiation pressure, the thickness of the weak portion arranged in the first region can be larger. In this way, when the battery cell is normally used, the weak portion has a stronger ability to resist external impact, which is helpful to reduce the risk of the weak portion being damaged prematurely.

[0045] In a second aspect, an embodiment of the present application further provides a battery, including the above battery cell.

[0046] In a third aspect, an embodiment of the present application further provides an electrical device, including the above battery, where the battery is configured to provide electrical energy for the electrical device.

Description

DESCRIPTION OF THE DRAWINGS

[0047] To more clearly describe the technical solutions of the embodiments of the present application, the drawings to be used in the embodiments will be briefly introduced below, and it should be understood that the following drawings only show some embodiments of the present application, and therefore should not be considered as limiting the scope of the present application. For those of ordinary skills in the art, other relevant drawings may also be obtained based on these drawings without creative efforts.

[0048] FIG. 1 is a schematic structural diagram of a vehicle provided in some embodiments of the present application;

[0049] FIG. 2 is an exploded view of a battery provided by some embodiments of the present application;

[0050] FIG. 3 is a schematic structural view of a battery cell according to some embodiments of the present application;

[0051] FIG. 4 is a top view of the battery cell according to some embodiments of the present application;

[0052] FIG. 5 is a schematic structural view of a battery cell according to some further embodiments of the present application;

[0053] FIG. 6 is a top view of the battery cell according to some further embodiments of the present application;

[0054] FIG. 7 is a top view of a battery cell according to some other embodiments of the present application;

[0055] FIG. 8 is a sectional view taken along E-E in FIG. 7.

[0056] FIG. 9 is a top view of a battery cell according to yet some other embodiments of the present application;

[0057] FIG. 10 is a top view of a battery cell according to still some other embodiments of the present application;

[0058] FIG. 11 is a top view of a battery cell according to some further other embodiments of the present application;

[0059] FIG. 12 is a top view of a battery cell according to yet some further other embodiments of the present application;

[0060] FIG. 13 is a top view of a battery cell according to still some further other embodiments of the present application;

[0061] FIG. 14 is a top view of a battery cell according to yet still some further embodiments of the present application;

[0062] FIG. 15 is a side view of a battery cell according to yet still some other embodiments of the present application; and

[0063] FIG. 16 is a schematic structural view of a battery cell according to yet still some further other embodiments of the present application.

[0064] Reference numerals: **10**—box body; **11**—first portion; **12**—second portion; **20**—battery cell; **21**—weak portion; **211**—first weak section; **2111**—weak point; **212**—second weak section; **2121**—first end; **2122**—second end; **213**—third weak section; **2131**—third end; **2132**—fourth end; **214**—fourth weak section; **215**—fifth weak section; **216**—sixth weak section; **22**—shell; **221**—wall portion; **2211**—first region; **2212**—score groove; **2213**—center point; **222**—case; **2221**—side wall; **2222**—bottom wall; **223**—end cover; **23**—electrode terminal; **100**—battery; **200**—controller; **300**—motor; **1000**—vehicle.

DETAILED DESCRIPTION

[0065] In order to make the objects, technical solutions and advantages of embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be clearly described below with reference to the drawings for the embodiments of the present application. Apparently, the described embodiments are some of, rather than all of, the embodiments of the present application. All the other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present application without any creative effort shall fall within the scope of protection of the present application.

[0066] Unless otherwise defined, all technical and scientific terms used in the present application shall have the same meanings as those generally understood by those skilled in the art of the present application. The terms used in the present application in the specification of application are merely for the purpose of describing specific embodiments and are not intended to limit the present application. The terms “include” and “have” and any variations thereof in the specification and claims and the above brief description of the drawings of the present application are intended to

cover non-exclusive inclusion. The terms “first,” “second,” etc. in the specification and the claims of the present application as well as the above drawings are used to distinguish different objects, rather than to describe a specific order or primary-secondary relationship.

[0067] The phrase “embodiment” referred to in the present application means that the descriptions of specific features, structures, and characteristics in combination with the embodiment are included in at least one embodiment of the present application. The appearance of this phrase in various places in the specification does not necessarily refer to the same embodiment, nor is it a separate or alternative embodiment that is mutually exclusive with other embodiments.

[0068] In the description of the present application, it should be noted that the terms “mounting,” “connecting,” “connection” and “attachment” should be understood in a broad sense, unless otherwise explicitly specified or defined, for example, it may be a fixed connection, a detachable connection or an integrated connection; and may be a direct connection or an indirect connection through an intermediate medium, or may be a communication between the interior of two elements. For those of ordinary skill in the art, the specific meanings of the above terms in the present application can be understood according to specific situations.

[0069] In the present application, the term “and/or” is only an association relation describing associated objects, which means that there may be three relations, for example, A and/or B may represent three situations: A exists alone, both A and B exist, and B exists alone. In addition, the character “/” in the present application generally means that the associated objects before and after it are in an “or” relationship.

[0070] In the embodiments of the present application, the same reference signs denote the same components, and for the sake of brevity, detailed descriptions of the same components are omitted in different embodiments. It should be understood that the thickness, length, width and other dimensions of the various components in the embodiments of the present application shown in the drawings, as well as the overall thickness, length, width and other dimensions of an integrated apparatus, are for illustrative purposes only, and should not constitute any limitation to the present application.

[0071] In the present application, the phrase “a plurality of” means two or more.

[0072] In the present application, the battery cell may include a lithium-ion secondary battery, a lithium-ion primary battery, a lithium sulfur battery, a sodium lithium-ion battery, a sodium-ion battery or a magnesium-ion battery and so on. This is not limited in the embodiments of the present application. The battery cell may be in a cylindrical shape, a flat shape, a cuboid shape or another shape, which is also not limited in the embodiments of the present application. The battery cells are generally classified into three types depending on the way of encapsulation: cylindrical battery cells, prismatic battery cells and pouch battery cells, which are also not limited in the embodiments of the present application.

[0073] The battery mentioned in the embodiments of the present application refers to a single physical module including one or more battery cells to provide a higher voltage and capacity. For example, the battery mentioned in the present application may include a battery module, a battery pack, or the like. The battery generally includes a box for packaging one or more battery cells. The box can prevent liquid or other foreign matters from affecting charging or discharging of the battery cells.

[0074] The battery cell includes an electrode assembly and an electrolytic solution, and the electrode assembly is composed of a positive electrode plate, a negative electrode plate, and a separator. The operation of the battery cell mainly relies on the movement of metal ions between the positive electrode plate and the negative electrode plate. The positive electrode plate includes a positive electrode current collector and a positive electrode active material layer, a surface of the positive electrode current collector is coated with the positive electrode active material layer, the positive electrode current collector not coated with the positive electrode active material layer protrudes from the positive electrode current collector already coated with the positive electrode

active material layer, and the positive electrode current collector not coated with the positive electrode active material layer is used as a positive tab. Taking a lithium-ion battery as an example, the material of the positive electrode current collector may be aluminum, and the positive electrode active material may be lithium cobalt oxide, lithium iron phosphate, ternary lithium, lithium manganate, or the like. The negative electrode plate includes a negative electrode current collector and a negative electrode active material layer, a surface of the negative electrode current collector is coated with the negative electrode active material layer, the negative electrode current collector not coated with the negative electrode active material layer protrudes from the negative electrode current collector already coated with the negative electrode active material layer, and the negative electrode current collector not coated with the negative electrode active material layer is used as a negative tab. The material of the negative electrode current collector may be copper, and the negative electrode active material may be, e.g., carbon or silicon. In order to ensure that no fusing occurs when a large current passes, there are a plurality of positive tabs which are stacked together, and there are a plurality of negative tabs which are stacked together. The material of the separator may be, e.g., polypropylene (PP) or polyethylene (PE). In addition, the electrode assembly may have a winded structure or a stacked structure, but the embodiments of the present application are not limited thereto.

[0075] For development of a battery technology, design factors of many aspects should be considered at the same time, such as the energy density, the cycle life, the discharging capacity, the charging-discharging rate, and other performance parameters. In addition, the reliability of the battery should also be considered.

[0076] For a battery cell, in order to ensure the reliability of the battery cell, a shell of the battery cell may be provided with a pressure relief mechanism. For example, an end cover is provided with a weak portion. When the internal pressure of the battery cell reaches a fracture initiation pressure, the weak portion is destroyed to relieve the internal pressure of the battery cell to reduce the risk of explosion or fire of the battery cell. However, the weak portion is often opened prematurely, causing the normal pressure relief function to fail.

[0077] The shell of the battery cell is easily subjected to external impact, which can easily act on the weak portion, and then the weak portion is opened prematurely, causing the normal pressure relief function to fail.

[0078] In view of this, embodiments of the present application provide a battery cell, where the battery cell includes a shell, the shell has a wall portion, the wall portion includes a weak portion, and the weak portion is configured to be destroyed when the battery cell releases internal pressure. An outer surface of the wall portion has a center point. A maximum distance between a projection of the weak portion on the outer surface of the wall portion and the center point is a , the minimum radial dimension of the outer surface of the wall portion is A , and a and A satisfy: $a \leq 0.4A$. The radial direction refers to a direction passing through the center point.

[0079] By setting $a \leq 0.4A$, namely, the maximum distance between the projection of the weak portion on the outer surface of the wall portion and the center point of the outer surface of the wall portion is less than or equal to 0.4 times the minimum radial dimension of the outer surface of the wall portion, the weak portion is arranged in a first region, with a relatively small stiffness, of the wall portion. The first region is a circular region with the center point of the outer surface of the wall portion as a circle center and 0.4 times the minimum radial dimension of the outer surface of the wall portion as a radius. The first region is close to the center of the wall portion, and the first region is a region, with a relatively small stiffness, of the wall portion and thus has a relatively weak ability to resist deformation. When the battery cell releases the internal pressure, the first region undergoes a large deformation under the action of gas, and therefore the weak portion arranged in the first region is easily deformed and damaged. Therefore, under the condition of the same fracture initiation pressure, the thickness of the weak portion arranged in the first region can be larger. In this way, when the battery cell is normally used, the weak portion has a stronger ability

to resist external impact, which is helpful to reduce the risk of the weak portion being damaged prematurely. Furthermore, the thicker the weak portion is, the easier it is to manufacture, and the lower the precision requirement for a manufacturing device.

[0080] The technical solutions described in the embodiments of the present application are applicable to batteries and electrical devices using the batteries.

[0081] The electrical device may be a vehicle, a mobile phone, a portable device, a laptop, a ship, a spacecraft, an electric toy, an electric tool, etc. The spacecraft includes an airplane, a rocket, a space shuttle, or a spaceship. The electric toy includes a fixed or mobile electric toy, such as a game console, an electric vehicle toy, an electric ship toy, and an electric airplane toy. The electric tool includes an electric tool for metal cutting, an electric tool for grinding, an electric tool for assembling and an electric tool for railways, such as an electric drill, an electric grinder, an electric wrench, an electric screwdriver, an electric hammer, an electric impact drill, a concrete vibrator, and an electric planer. The electrical device is not specially limited in the embodiments of the present application.

[0082] For the convenience of illustration, the following embodiments are illustrated with the electrical device being a vehicle **1000** as an example.

[0083] Referring to FIG. 1, FIG. 1 is a schematic structural diagram of a vehicle **1000** provided in some embodiments of the present application. The vehicle **1000** may be a fuel vehicle, a gas vehicle or a new energy vehicle. The new energy vehicle may be an all-electric vehicle, a hybrid vehicle, an extended-range vehicle, or the like. A battery **100** is arranged in the vehicle **1000**. The battery **100** may be arranged at the bottom or head portion or tail portion of the vehicle **1000**. The battery **100** may be used as a power supply for the vehicle **1000**, for example, the battery **100** may be used as an operating power source for the vehicle **1000**. The vehicle **1000** may further include a controller **200** and a motor **300**. The controller **200** is configured to control the battery **100** to supply power to the motor **300**, for example, to supply power for starting, navigation and driving of the vehicle **1000**.

[0084] In some embodiments of the present application, the battery **100** not only may serve as an operating power source of the vehicle **1000**, but also may serve as a driving power source of the vehicle **1000**, thus replacing or partially replacing fuel or natural gas to provide driving power for the vehicle **1000**.

[0085] Referring to FIG. 2, FIG. 2 is an exploded view of a battery **100** provided in some embodiments of the present application. The battery **100** comprises a box body **10** and a battery cell **20**, wherein the battery cell **20** is accommodated within the box body **10**. The box body **10** is used to provide an accommodating space for the battery cell **20**, and the box body **10** can be of various structures. In some embodiments, the box body **10** may include a first part **11** and a second part **12**, the first part **11** and the second part **12** covering each other, and the first part **11** and the second part **12** together defining the accommodating space for accommodating the battery cell **20**. The second part **12** may be of a hollow structure with an opening at one end, and the first part **11** may be of a plate-like structure, where the first part **11** covers the opening side of the second part **12** so that the first part **11** and the second part **12** together define the accommodation space. The first part **11** and the second part **12** may each be of a hollow structure with an opening at one end, where the opening side of the first part **11** covers the opening side of the second part **12**. Of course, the box body **10** formed by the first part **11** and the second part **12** may be of a variety of shapes, such as a cylinder and a rectangular solid.

[0086] In the battery **100**, there may be a plurality of battery cells **20**, and the plurality of battery cells **20** may be connected in series or in parallel or in parallel-series, where the parallel-series connection means that the plurality of battery cells **20** are connected in both series and parallel. The plurality of battery cells **20** may be directly connected together in series or in parallel or in parallel-series, and then the whole composed of the plurality of battery cells **20** may be accommodated in the box body **10**. Of course, the battery **100** may also be in the form of a battery module first

formed by the plurality of battery cells **20** being connected in series or in parallel or in parallel-series, then the plurality of battery modules may be connected in series or in parallel or in parallel-series to form a whole, and accommodated in the box body **10**. The battery **100** may further include other structures. For example, the battery **100** may further include a bus component for electrically connecting the plurality of battery cells **20**.

[0087] Each battery cell **20** may be a secondary battery cell or a primary battery cell; or may be a lithium-sulfur battery cell, a sodium-ion battery cell or a magnesium-ion battery cell, but is not limited thereto. The battery cell **20** can be in a cylinder shape, a flat body shape, a cuboid shape or other shapes.

[0088] Referring to FIG. 3, FIG. 4, FIG. 5, and FIG. 6, FIG. 3 is a schematic structural view of a battery cell **20** according to some embodiments of the present application. FIG. 4 is a top view of the battery cell **20** according to some embodiments of the present application. FIG. 5 is a schematic structural view of a battery cell **20** according to some further embodiments of the present application. FIG. 6 is a top view of the battery cell **20** according to some further embodiments of the present application. Embodiments of the present application provides a battery cell **20**, where the battery cell **20** includes a shell **22**, the shell **22** has a wall portion **221**, the wall portion **221** includes a weak portion **21**, and the weak portion **21** is configured to be destroyed when the battery cell **20** releases internal pressure. An outer surface of the wall portion **221** has a center point **2213**. A maximum distance between a projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** is a , the minimum radial dimension of the outer surface of the wall portion **221** is A , and a and A satisfy: $a \leq 0.4A$. The radial direction refers to a direction passing through the center point **2213**.

[0089] The shell **22** includes an end cover **223** and a case **222**. The case **222** has an accommodating space with one end open, and the accommodating space is configured to accommodate an electrode assembly. The end cover **223** is connected to the case **222** and configured to close the opening.

[0090] The end cover **223** is a component that covers the opening of the case **222** to isolate an internal environment of the battery cell **20** from an external environment. Without limitation, a shape of the end cover **223** may be adaptive to that of the case **222** so as to fit the case **222**. Optionally, the end cover **223** may be made of a material (such as an aluminum alloy) with a certain hardness and strength, so that the end cover **223** is not likely to be deformed when being pressed and collided, the battery cell **20** can have a higher structural strength, and the reliability can also be improved. The end cover **223** may also be made of various materials, such as copper, iron, aluminum, stainless steel, an aluminum alloy, and plastic. This is not specially limited in the embodiments of the present application. In some embodiments, the battery cell **20** further includes an insulating member, the insulating member is arranged on an inner side of the end cover **223**, and the insulating member may be configured to isolate an electrical connection component in the case **222** from the end cover **223**, thereby reducing the risk of a short circuit. For example, the insulating member may be made of plastic, rubber, and the like.

[0091] The case **222** is a component configured to fit the end cover **223** to form the internal environment of the battery cell **20**, where the formed internal environment may be configured to accommodate an electrode assembly **23**, an electrolytic solution, and other components. The case **222** and the end cover **223** may be separate components, the opening may be formed in the case **222**, and at the opening, the opening is covered with the end cover **223** so as to form the internal environment of the battery cell **20**. Without limitation, the end cover **223** and the case **222** may also be integrated. Specifically, the end cover **223** and the case **222** may first form a common connection surface before other components enter the case. When the inside of the case **222** is required to be encapsulated, the case **222** is covered with the end cover **223**. The case **222** may have various shapes and sizes, such as a cuboid shape, a cylinder shape, and a hexagonal prism shape. Specifically, the shape of the case **222** may be determined according to the specific shape and size of the electrode assembly **23**. The case **222** may be made of various materials, such as

copper, iron, aluminum, stainless steel, an aluminum alloy, and plastic, which is not particularly limited in the embodiments of the present application.

[0092] The electrode assembly is a component where an electrochemical reaction occurs in the battery cell **20**. One or more electrode assemblies may be contained within the shell **22**. The electrode assembly is mainly formed by winding or stacking a positive electrode plate and a negative electrode plate, and a separator is generally arranged between the positive electrode plate and the negative electrode plate. The portions of the positive electrode plate and the negative electrode plate that have active materials constitute a main body portion of the electrode assembly, and the portions of the positive electrode plate and the negative electrode plate that do not have active materials each constitute a tab. A positive electrode tab and a negative electrode tab may be co-located at one end of the main body portion or located at two ends of the main body portion, respectively. During charging and discharging of the battery **100**, a positive electrode active material and a negative electrode active material react with the electrolyte solution.

[0093] An electrode terminal **23** may be arranged on the shell **22** and the electrode terminal **23** may be configured to be electrically connected to the electrode assembly, for outputting or inputting electrical energy of the battery cell **20**. The electrode terminal **23** may be arranged on the end cover **223**, or on the case **222**. In the embodiments shown in FIG. 3, the electrode terminal **23** is arranged on the end cover **223**.

[0094] The weak portion **21** is a structure which is configured to be opened to release the internal pressure of the battery cell **20** and reduce the risk of explosion or fire of the battery cell **20** when the internal pressure or temperature of the battery cell **20** reaches the fracture initiation pressure.

[0095] The shell **22** has a plurality of walls, such as a bottom wall **2222**, side walls **2221**, etc. Each wall of the shell **22** may serve as the wall portion **221**. For example, the bottom wall **2222** may be the wall portion **221**, and in this case, the weak portion **21** is arranged on the bottom wall **2222**. For another example, the side wall **2221** may be the wall portion **221**, and in this case, the weak portion **21** is arranged on the side wall **2221**. Particularly, the end cover **223** can be used alone as a wall portion **221**, and in this case, the weak portion **21** is arranged on the end cover **223**. It should be noted that the shell **22** may have a plurality of wall portions **221**, and in this case, each wall portion **221** is provided with the weak portion **21**. For example, the bottom wall **2222** and the side wall **2221** are both the wall portions **221**, and in this case, the bottom wall **2222** and the side wall **2221** are both provided with the weak portion **21**.

[0096] An outer surface of the wall portion **221** has a center point **2213**. Referring to FIG. 3 and FIG. 4, taking the wall portion **221** as a rectangular structure as an example, the center point **2213** of the outer surface of the wall portion **221** is a point of intersection of diagonal lines of the rectangle. Referring to FIG. 5 and FIG. 6, taking the wall portion **221** as a circular structure as an example, the center point **2213** of the outer surface of the wall portion **221** is the center of the circle.

[0097] A represents the maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213**, namely, a distance between a point, farthest from the center point **2213**, in the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213**. For example, the weak portion **21** is circular, and the center of the circle coincides with the center point **2213**. In this case, the maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** is the radius of the circle corresponding to the weak portion **21**. For another example, the weak portion **21** is circular, any point on the weak portion **21** coincides with the center point **2213**. In this case, the maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** is the diameter of the circle corresponding to the weak portion **21**.

[0098] A represents the minimum radial dimension of the outer surface of the wall portion **221**. “The minimum radial dimension of the outer surface of the wall portion **221**” refers to the

dimension of a shortest linear segment among a plurality of linear segments that have any two points on the edge of the wall portion **221** as endpoints and pass through the center point **2213**. Referring to FIG. 3 and FIG. 4, in the embodiments shown in FIG. 3 and FIG. 4, the outer surface of the wall portion **221** is in the shape of a rectangle, and the minimum radial dimension of the outer surface of the wall portion **221** is the width of the rectangle. Referring to FIG. 5 and FIG. 6, in the embodiments shown in FIG. 5 and FIG. 6, the outer surface of the wall portion **221** is in the shape of a circle, and the minimum radial dimension of the outer surface of the wall portion **221** is the diameter of the circle. For another example, in some embodiments, the outer surface of the wall portion **221** is in the shape of a regular hexagon, and the minimum radial dimension of the outer surface of the wall portion **221** is a distance between two parallel edges of the regular hexagon.

[0099] “The maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** is a , the minimum radial dimension of the outer surface of the wall portion **221** is A , and a and A satisfy: $a \leq 0.4A$ ” can be understood as: the weak portion **21** is located in the first region **2211** of the wall portion **221**, and the edge of the weak portion **21** can overlap with the edge of the first region **2211**, but cannot exceed the first region **2211**. The first region **2211** is a circular region with the center point **2213** of the outer surface of the wall portion **221** as a circle center and 0.4 times the minimum radial dimension of the outer surface of the wall portion **221** as a radius. The first region **2211** is a region, close to the center of the wall portion **221**, of the wall portion **221**. Compared with a region close to the edge of the wall portion **221**, the first region **2211** has a smaller stiffness. When the internal pressure of the battery cell **20** reaches the fracture initiation pressure, the first region **2211** has a large deformation.

[0100] Referring to FIG. 3, FIG. 4, FIG. 5, and FIG. 6, in FIG. 3, FIG. 4, FIG. 5, and FIG. 6, the boundary of the first region **2211** is shown by a dotted line, and the first region **2211** is a region defined by the dotted line. It should be noted that the dotted line is only used to show the boundary of the first region **2211** and does not mean that there are other physical structures blocked at the location of the dotted line.

[0101] By setting $a \leq 0.4A$, namely, the maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** of the outer surface of the wall portion **221** is less than or equal to 0.4 times the minimum radial dimension of the outer surface of the wall portion **221**, the weak portion **21** is arranged in the first region **2211**, with a relatively small stiffness, of the wall portion **221**. The first region **2211** is a circular region with the center point **2213** of the outer surface of the wall portion **221** as a circle center and 0.4 times the minimum radial dimension of the outer surface of the wall portion **221** as a radius. The first region **2211** is close to the center of the wall portion **221**, and the first region **2211** is a region, with a relatively small stiffness, of the wall portion **221** and thus has a relatively weak ability to resist deformation. When the battery cell **20** releases the internal pressure, the first region **2211** undergoes a large deformation under the action of gas, and therefore the weak portion **21** arranged in the first region **2211** is easily deformed and damaged. Therefore, under the condition of the same fracture initiation pressure, the thickness of the weak portion **21** arranged in the first region **2211** can be larger. In this way, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, which is beneficial to reduce the risk of the weak portion **21** being damaged prematurely. Furthermore, the thicker the weak portion **21** is, the easier it is to manufacture, and the lower the precision requirement for a manufacturing device.

[0102] Referring to FIG. 7 and FIG. 8, FIG. 7 is a top view of a battery cell **20** according to some other embodiments of the present application. FIG. 8 is a sectional view taken along E-E in FIG. 7. In some embodiments, the wall portion **221** is provided with a score groove **2212**, and the weak region **21** is formed at a bottom of the score groove **2212**.

[0103] The score groove **2212** may be formed in a variety of ways, such as stamping, milling, etc. The score groove **2212** may be formed in a surface of the wall portion **221** facing the inside of the shell **22**, or may be formed in a surface of the wall portion **221** facing away from the inside of the

shell **22**. Taking the wall portion **221** of a rectangular flat plate structure as an example, the wall portion **221** has opposite inner and outer surfaces in a thickness direction. The inner surface of the wall portion **221** faces the inside of the shell **22**, and the outer surface of the wall portion **221** faces away from the inside of the shell **22**. The score groove **2212** may be formed in the inner surface of the wall portion **221** or formed in the outer surface of the wall portion **221**.

[0104] The score groove **2212** may be a groove recessed from the outer surface of the wall portion **221** in the thickness direction of the wall portion **221**, and the weak portion **21** is a part, between the inner surface thereof and a bottom surface of the score groove **2212**, of the wall portion **221**.

The score groove **2212** may also be a groove recessed from the inner surface of the wall portion **221** in the thickness direction of the wall portion **221**, and the weak portion **21** is a part, between the outer surface thereof and the bottom surface of the score groove **2212**, of the wall portion **221**.

[0105] The score groove **2212** is described as a groove sunken from the outer surface of the wall portion **221** in the thickness direction of the wall portion **221**. Referring to FIG. 3 and FIG. 4, the score groove **2212** may be a linear groove extending along a linear trajectory. Referring to FIG. 5 and FIG. 6, the score groove **2212** may also be an annular groove extending along a closed trajectory.

[0106] The weak portion **21** is formed by forming the score groove **2212** in the wall portion **221**, with operation simple and convenient and costs low.

[0107] Referring to FIG. 7 and FIG. 8, in some embodiments, the projection of the weak portion **21** on the outer surface of the wall portion **221** passes through the center point **2213**.

[0108] “The projection of the weak portion **21** on the outer surface of the wall portion **221** passes through the center point **2213**” may be understood as: the projection of the weak portion **21** on the outer surface of the wall portion **221** covers the center point **2213**. The center point **2213** may be located at the edge of the projection of the weak portion **21** on the outer surface of the wall portion **221**. For example, the weak portion **21** is of a linear structure, and the center point **2213** is located at one end of the projection of the weak portion **21** on the outer surface of the wall portion **221**. The center point **2213** may be located at the middle of the projection of the weak portion **21** on the outer surface of the wall portion **221**. For example, the weak portion **21** is of a linear structure, and the center point **2213** is located at a midpoint of the projection of the weak portion **21** on the outer surface of the wall portion **221**.

[0109] When the battery cell **20** releases the internal pressure, the center point **2213** is a position, with the largest deformation, of the wall portion **221** and is most likely to be damaged. By making the projection of the weak portion **21** on the outer surface of the wall portion **221** pass through the center point **2213**, the thickness of the weak portion **21** at the center point **2213** can be larger. In this way, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely.

[0110] Referring to FIG. 7 and FIG. 8, in some embodiments, the weak portion **21** has a weak point **2111** arranged corresponding to the center point **2213**, and the thickness of the weak portion **21** at the weak point **2111** is a first thickness. The maximum thickness of the weak portion **21** at other positions than the weak point **2111** is a second thickness. The first thickness is greater than or equal to the second thickness.

[0111] The first thickness is the thickness of the weak portion **21** at the weak point **2111**, or the thickness of the weak portion **21** at the location of the center point **2213**. Referring to FIG. 8, H.sub.1 is indicated as the first thickness in FIG. 8.

[0112] The second thickness is the maximum thickness of the weak portion **21** at other positions than the weak point **2111**. Referring to FIG. 8, if H.sub.3 in FIG. 8 is the maximum thickness of the weak portion **21** at other positions than the weak point **2111**, H.sub.3 represents the second thickness. Then: $H_{\text{sub.1}} \geq H_{\text{sub.3}}$.

[0113] In the embodiments shown in FIG. 8, the first thickness is greater than the second thickness, $H_{\text{sub.1}} > H_{\text{sub.3}}$. In some other embodiments, the first thickness may be equal to the second

thickness, H.sub.1=H.sub.3.

[0114] The first thickness is greater than the second thickness, namely, the thickness of the weak portion **21** at the weak point **211** is greater than the maximum thickness of the weak portion **21** at other positions than the weak point **211**. The thickness of the weak portion **21** at the weak point **211** is greater than the thickness of the weak portion **21** at other positions, and the weak point **211** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely. The first thickness is equal to the second thickness, namely, the thickness of the weak portion **21** at the weak point **211** is equal to the maximum thickness of the weak portion **21** at other positions than the weak point **211**. In this way, the thickness of the weak portion **21** is relatively more uniform, thereby facilitating processing and manufacturing.

[0115] Referring to FIG. 7 and FIG. 8, in some embodiments, the score groove **2212** includes at least one groove section, the weak portion **21** includes at least one weak section, and the at least one groove section and the at least one weak section are arranged in a one-to-one correspondence.

[0116] The score groove **2212** may include two groove sections, and the two groove sections are cross-arranged. Correspondingly, two weak sections are arranged on the weak portion **21**, and the two weak sections are cross-arranged.

[0117] Referring to FIG. 7 and FIG. 8, the score groove **2212** includes three groove sections. Correspondingly, three weak sections are arranged on the weak portion **21**. An arrangement form of the three weak sections is the same as that of the three groove sections.

[0118] Of course, the score groove **2212** may include more than three groove sections. Correspondingly, more than three weak sections are arranged on the weak portion **21**. An arrangement form of the more than three weak sections is the same as that of the more than three groove sections.

[0119] Each time one groove section is arranged on the wall portion **221**, one weak section is formed correspondingly. During manufacturing, quick forming can be achieved by stamping or milling, with operations simple and convenient and costs low.

[0120] Referring to FIG. 7 and FIG. 8, in some embodiments, the at least one weak section includes at least one first weak section **211**, and the weak point **211** is arranged on the first weak section **211**. The thickness of the first weak section **211** at the weak point **211** is the first thickness. The thickness of the first weak section **211** at other positions than the weak point **211** is the third thickness. The first thickness is greater than or equal to the third thickness.

[0121] The at least one weak section may include one first weak section **211**, two first weak sections **211**, three first weak sections **211**, and more than three first weak sections **211**. When the at least one weak section includes a plurality of first weak sections **211**, the plurality of weak sections all intersect at the weak point **211**.

[0122] Referring to FIG. 7 and FIG. 8, in the embodiments shown in FIG. 7 and FIG. 8, the at least one weak section includes one first weak section **211**. The thickness of the first weak section **211** at the weak point **211** is the first thickness, as indicated by H.sub.1 in FIG. 8.

[0123] The thickness of the first weak section **211** at other positions than the weak point **211** is the third thickness. The thickness of the first weak section **211** at other positions than the weak point **211** may be non-uniform. For example, the thickness of the first weak section **211** at other positions than the weak point **211** may gradually increase or decrease in an extension direction of the first weak section **211**. In this case, the third thickness is the maximum thickness of the weak portion **21** at other positions than the weak point **211**. The thickness of the first weak section **211** at other positions than the weak point **211** may be uniform, namely, the thickness of the first weak section **211** at any other position than the weak point **211** is the same, and the thickness of the first weak section **211** at any other position than the weak point may be taken as the third thickness.

[0124] Referring to FIG. 8, the third thickness is indicated by H.sub.3 in FIG. 8. In the embodiments shown in FIG. 8, the thickness of the first weak section **211** at the weak point **211** is greater than the thickness of the first weak section **211** at other positions than the weak point **211**,

namely, $H_{\text{sub.1}} > H_{\text{sub.3}}$.

[0125] In some other embodiments, the thickness of the first weak section **211** at the weak point **2111** may be equal to the thickness of the first weak section **211** at other positions than the weak point **2111**, namely, $H_{\text{sub.1}} = H_{\text{sub.3}}$. In this case, the thickness of the first weak section **211** at other positions than the weak point **2111** represents both the third thickness and the second thickness.

[0126] The first thickness is greater than the third thickness, namely, the thickness of the first weak section **211** at the weak point **2111** is greater than the thickness of the first weak section **211** at other positions than the weak point **2111**. The thickness of the first weak section **211** at the weak point **2111** is greater than the thickness of the first weak section at other positions, and therefore the weak point **2111** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely. The first thickness is equal to the third thickness, namely, the thickness of the first weak section at the weak point **2111** is equal to the thickness of the first weak section **211** at other positions than the weak point **2111**. In this way, the thickness of the first weak section **211** is relatively more uniform, thereby facilitating processing and manufacturing.

[0127] Referring to FIG. 7 and FIG. 8, in some embodiments, the at least one weak section further includes a second weak section **212** and a third weak section **213**. The second weak section **212** and the third weak section **213** are arranged opposite to each other and spaced apart. The first weak section **211** is configured to connect the second weak section **212** and the third weak section **213**. The thickness of the second weak section **212** and the thickness of the third weak section **213** are both the fourth thickness, and the fourth thickness is smaller than the third thickness.

[0128] The first weak section **211**, the second weak section **212**, and the third weak section **213** may all be linear segments, or may all be non-linear segments, such as circular arc-shaped sections. In an embodiment where the first weak section **211**, the second weak section **212**, and the third weak section **213** are all linear segments, it can be understood that the first weak section **211**, the second weak section **212**, and the third weak section **213** all extend along a straight line, and the second weak section **212** and the third weak section **213** may be arranged in parallel or at an angle. The second weak section **212** and the third weak section **213** may both be perpendicular to the first weak section **211**, or may both be non-perpendicular to the first weak section **211**.

[0129] A connection position between the second weak section **212** and the first weak section **211** may be located at one end of the second weak section **212**, or at a position deviating from one end of the second weak section **212**. For example, the connection position between the second weak section **212** and the first weak section **211** is located at a midpoint of the second weak section **212** in an extension direction. A connection position between the third weak section **213** and the first weak section **211** may be located at one end of the third weak section **213**, or at a position deviating from one end of the third weak section **213**. For example, the connection position between the third weak section **213** and the first weak section **211** is located at a midpoint of the third weak section **213** in an extension direction.

[0130] The thickness of the second weak section **212** is the same as the thickness of the third weak section **213**. The thickness of the second weak section **212** may be non-uniform. For example, the thickness of the second weak section **212** may gradually increase in the extension direction thereof. In this case, the fourth thickness is the maximum thickness of the second weak section **212**. The thickness of the second weak section **212** may be uniform, namely, the thickness of the second weak section **212** at any position is the same, and the thickness of the second weak section **212** at any position may be taken as the fourth thickness.

[0131] Referring to FIG. 8, the fourth thickness is as indicated by $H_{\text{sub.4}}$ in FIG. 8. The fourth thickness is less than the third thickness, namely, $H_{\text{sub.4}} < H_{\text{sub.3}}$. In this case, the maximum thickness of the second weak section **212** is less than the minimum thickness of the first weak section **211** at any other position than the weak point.

[0132] When the battery cell **20** releases the pressure, the weak portion **21** can be opened along the

first weak section **211**, the second weak section **212**, and the third weak section **213**, forming a larger pressure relief area, which is beneficial to improve a pressure relief rate. The first weak section **211** is closer to the center of the first region **2211** than the second weak section **212** and the third weak section **213**. The thickness of the first weak section **211** at other positions than the weak point **2111** may also be greater than the thickness of the second weak section **212** and the third weak section **213**, thereby improving the ability of the weak portion **21** to resist external impact and reducing the risk of the weak portion **21** being damaged prematurely.

[0133] Referring to FIG. 7 and FIG. 8, in some embodiments, one end of the first weak section **211** is connected to the middle of the second weak section **212**, and the other end of the first weak section **211** is connected to the middle of the third weak section **213**. The weak point **2111** is located at the midpoint of the first weak section **211**.

[0134] The first weak section **211** has two opposite ends, where one end is connected to the second weak section **212**, and the other end is connected to the third weak section **213**.

[0135] “One end of the first weak section **211** is connected to the middle of the second weak section **212**” means that one end of the first weak section **211** is connected to the midpoint of the second weak section **212** or a region within 3 mm from the midpoint.

[0136] “The other end of the first weak section **211** is connected to the middle of the third weak section **213**” means that the other end of the first weak section **211** is connected to the midpoint of the third weak section **213** or a region within 3 mm from the midpoint.

[0137] Referring to FIG. 7, in the embodiments shown in FIG. 7, the first weak section **211**, the second weak section **212**, and the third weak section **213** form an H-shaped weak portion **21**. The midpoint of the first weak section **211** is the weak point **2111**.

[0138] When the battery cell **20** releases the pressure, after being destroyed, the weak point **2111** can rupture along the first weak section **211** toward the second weak section **212** and the third weak section **213**, respectively. After reaching a point of intersection of the first weak section **211** and the second weak section **212**, the weak point ruptures from the point of intersection in the extension direction of the second weak section **212** toward both ends of the second weak section **212**. After reaching a point of intersection of the first weak section **211** and the third weak section **213**, the weak point ruptures from the point of intersection in the extension direction of the third weak section **213** toward both ends of the third weak section **213** to achieve fast pressure relief.

[0139] Referring to FIG. 9, FIG. 9 is a top view of a battery cell **20** according to yet some other embodiments of the present application. In yet some other embodiments, the second weak section **212** includes a first end **2121** and a second end **2122**, the third weak section **213** includes a third end **2131** and a fourth end **2132**, the first end **2121** and the third end **2131** are arranged opposite to each other, and the second end **2122** and the fourth end **2132** are arranged opposite to each other. The first weak section **211** is configured to connect the first end **2121** and the fourth end **2132** or connect the second end **2122** and the third end **2131**. The weak point **2111** is located at the midpoint of the first weak section **211**.

[0140] The first end **2121** and the second end **2122** are two ends of the second weak section **212** in the extension direction thereof. The third end **2131** and the fourth end **2132** are two ends of the third weak section **213** in the extension direction thereof. The first end **2121** and the third end **2131** are opposite to each other, and the second end **2122** and the fourth end **2132** are opposite to each other.

[0141] The first weak section **211** is obliquely arranged, and one end of the first weak section **211** may be connected to the first end **2121**. In this case, the other end of the first weak section **211** is connected to the fourth end **2132**. One end of the first weak section **211** may be connected to the second end **2122**, and in this case, the other end of the first weak section **211** is connected to the third end **2131**. The first weak section **211**, the second weak section **212**, and the third weak section **213** form a Z-shaped score groove **2212**. The midpoint of the first weak section **211** is the weak point **2111**.

[0142] When the battery cell **20** releases the pressure, after being destroyed, the weak point **2111** can rupture along the first weak section **211** toward the second weak section **212** and the third weak section **213**, respectively. After reaching the point of intersection of the first weak section **211** and the second weak section **212**, the weak point ruptures from the point of intersection in the extension direction of the second weak section **212** toward the other end of the second weak section **212**. After reaching the point of intersection of the first weak section **211** and the third weak section **213**, the weak point ruptures from the point of intersection in the extension direction of the third weak section **213** toward the other end of the third weak section **213** to achieve fast pressure relief.

[0143] Referring to FIG. **10**, FIG. **10** is a top view of a battery cell **20** according to still some other embodiments of the present application. In still some other embodiments, the at least one weak section further includes a fourth weak section **214** and a fifth weak section **215**, and the fourth weak section **214** and the fifth weak section **215** are arranged opposite to each other and spaced apart. The second weak section **212** and the fourth weak section **214** intersect at the first end **2121**, and the third weak section **213** and the fifth weak section **215** intersect at the second end **2122**. The first weak section **211** is configured to connect the first end **2121** and the second end **2122**, and the weak point **2111** is located at the midpoint of the first weak section **211**.

[0144] The first weak section **211**, the second weak section **212**, the third weak section **213**, the fourth weak section **214**, and the fifth weak section **215** may all be linear segments, or non-linear segments, such as circular arc-shaped grooves. The first weak section **211**, the second weak section **212**, and the fourth weak section **214** intersect at the first end **2121**. The first weak section **211**, the third weak section **213**, and the fifth weak section **215** intersect at the second end **2122**. The first end **2121** and the second end **2122** are two ends of the first weak section **211** in the extension direction thereof.

[0145] When the battery cell **20** release the pressure, after being destroyed, the weak point **2111** can rupture along the first weak section **211** toward the first end **2121** and the second end **2122**, respectively. After reaching the first end **2121**, the weak point ruptures from the first end **2121** along the second weak section **212** and the fourth weak section **214**, respectively. After reaching the second end **2122**, the weak point ruptures from the second end **2122** along the third weak section **213** and the fourth weak section **214**, respectively, to achieve fast pressure relief.

[0146] Referring to FIG. **11**, FIG. **11** is a top view of a battery cell **20** according to some further other embodiments of the present application. In some further other embodiments, the at least one weak section further includes a sixth weak section **216**, and the first weak section **211** and the sixth weak section **216** intersect at the weak point **2111**.

[0147] The sixth weak section **216** may be a linear segment or a non-linear segment, such as a circular arc-shaped section. In an embodiment where the first weak section **211**, the second weak section **212**, the third weak section **213**, and the sixth weak section **216** are all linear segments, it can be understood that the first weak section **211**, the second weak section **212**, the third weak section **213**, and the sixth weak section **216** all extend along a straight line, and the second weak section **212** and the sixth weak section **216** may be arranged in parallel or at an angle. The third weak section **213** and the sixth weak section **216** may be arranged in parallel or at an angle. The first weak section **211** may be perpendicular to the sixth weak section **216** or may be non-perpendicular to the sixth weak section **216**.

[0148] A connection position between the first weak section **211** and the sixth weak section **216** may be located at one end of the sixth weak section **216**, or at a position deviating from one end of the sixth weak section **216**. For example, the connection position between the first weak section **211** and the sixth weak section **216** is located at a midpoint of the sixth weak section **216** in an extension direction.

[0149] When the weak point **2111** serves as a point of intersection of the first weak section **211** and the sixth weak section **216**, the weak point has a more concentrated stress and thus is more likely to rupture. Therefore, the thickness of the weak portion **21** at the weak point **2111** can be set larger, so

that when the battery cell **20** is normally used, the weak point **2111** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely.

[0150] Referring to FIG. **12**, FIG. **12** is a top view of a battery cell **20** according to yet some further other embodiments of the present application. In yet some further other embodiments, the projection of the weak portion **21** on the outer surface of the wall portion **221** is a closed structure, and the closed structure surrounds the center point **2213**.

[0151] The projection of the weak portion **21** on the outer surface of the wall portion **221** may be circular, elliptical, polygonal, etc.

[0152] Referring to FIG. **12**, in the embodiments shown in FIG. **12**, the projection of the weak portion **21** on the outer surface of the wall portion **221** is in a runway shape. An area defined by the runway shape is relatively large, so that the weak portion **21** can form a larger opening on the wall portion **221** when the pressure is released, so as to facilitate fast pressure release.

[0153] The weak portion **21** is configured as a closed structure extending along a closed trajectory, so that when the internal pressure of the battery cell **20** reaches the fracture initiation pressure, the internal pressure can push open the weak portion **21** from all sides, forming the larger opening and allowing faster and easier pressure relief.

[0154] Referring to FIG. **11** and FIG. **12**, in some embodiments, in a thickness direction of the wall portion **221**, the center of the projection of the weak portion **21** coincides with the center point **2213**.

[0155] In the thickness direction of the wall portion **221**, the center of the projection of the weak portion **21** may be located inside the projection of the weak portion **21** or outside the projection of the weak portion **21**. For example, in the thickness direction of the wall portion **221**, the projection of the weak portion **21** is in the shape of a linear segment, and a midpoint of the linear segment is the center of the projection of the weak portion **21**, and in this case, the center of the projection of the weak portion **21** is located inside the projection of the weak portion **21**. For another example, in the thickness direction of the wall portion **221**, the projection of the weak portion **21** is in the shape of a ring, and the center of the ring is the center of the projection of the weak portion **21**, and in this case, the center of the projection of the weak portion **21** is located outside the projection of the weak portion **21**.

[0156] In the thickness direction of the wall portion **221**, the center of the weak portion **21** may coincide with the center point **2213**, or may be spaced apart from the center point **2213**. Both solutions can make the center of the projection of the weak portion **21** in the thickness direction coincide with the center of the projection of the center point **2213** in the thickness direction.

[0157] When the weak point **2111** coincides with the center point **2213**, the weak point has the largest deformation when the battery cell **20** releases the pressure and thus is most likely to rupture. Therefore, the thickness of the weak portion **21** at the weak point **2111** can be set larger so that when the battery cell **20** is normally used, the weak point **2111** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely.

[0158] Referring to FIG. **13** and FIG. **14**, FIG. **13** is a top view of a battery cell **20** according to still some further other embodiments of the present application. FIG. **14** is a top view of a battery cell **20** according to yet still some further embodiments of the present application. In some embodiments, in the thickness direction of the wall portion **221**, the center of the projection of the weak portion **21** deviates from the center point **2213**.

[0159] “In the thickness direction of the wall portion **221**, the center of the projection of the weak portion **21** deviates from the center point **2213**” may also be understood as: in the thickness direction of the wall portion **221**, the center of the projection of the weak portion **21** does not overlap with the center point **2213**. In other words, in a direction perpendicular to the thickness direction of the wall portion **221**, there is a gap between the center of the projection of the weak portion **21** and the center point **2213**.

[0160] By making the center of the projection of the weak portion **21** in the thickness direction

deviate from the center point **2213**, it is beneficial to improve the creep resistance of the weak portion **21**.

[0161] Referring to Table 1, Table 1 shows the test data of creep resistance when the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** at different distances. In Table 1, B represents the distance at which the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213**.

[0162] This test uses 87203 square case battery cells with a capacity of 298 Ah, where the size of the battery cell **20** is: 87 mm×203 mm×112 mm, and the battery cell is a Ni50 system.

[0163] The specific test method is: press a blunt thorn head with a diameter of 20 mm on the center of the projection of the weak portion **21** in the thickness direction, and check a sinking distance of the weak portion **21** after standing for 7 days. The greater the sinking distance, the worse the creep resistance.

TABLE-US-00001 TABLE 1 Experiment B (unit: Sinking distance number mm) (unit: mm)

1	0	1.76	2	0.25	1.54	3	0.5	1.56	4	0.75	1.56	5	1	1.54	6	1.25	1.53	7	1.5	1.49	8	1.75	1.52	9	2	1.50	10	2.25
1.51	11	2.5	1.48	12	2.75	1.46	13	3	1.46	14	3.5	1.63	15	4	1.67	16	4.5	1.66	17	5	1.66							

[0164] Referring to Table 1, it can be seen from the table that when the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213**, the creep resistance of the weak portion **21** will be better than the creep resistance when the center of the projection of the weak portion **21** in the thickness direction coincides with the center point **2213**.

[0165] Referring to FIG. 13 and FIG. 14, in some embodiments, the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** by a distance B, which satisfies: $0 < B \leq 5$ mm.

[0166] The value of the distance where the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** can be: B=0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.5, 1.7, 1.9, 2, 2.2, 2.4, 2.6, 2.8, 3, 3.5, 4, 4.5, 5, etc.

[0167] It can be seen from Table 1 that when the center of the projection of the weak portion **21** in the thickness direction coincides with the center point **2213**, the sinking distance is 1.76 mm. When the distance at which the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** is in the range of 0-5 mm (excluding 0), the sinking distance does not exceed 1.7 mm.

[0168] Therefore, by limiting the distance at which the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** in the range of 0-5 mm (excluding 0), the weak portion **21** has better creep resistance.

[0169] In some embodiments, $0 < B \leq 3$ mm.

[0170] The value of the distance at which the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** can be B=0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, etc.

[0171] It can be seen from Table 1 that when $3 \text{ mm} < B \leq 5 \text{ mm}$, the sinking distance is greater than 1.6. When $0 < B \leq 3 \text{ mm}$, the sinking distance does not exceed 1.6. Therefore, when $0 < B \leq 3 \text{ mm}$, the weak portion **21** has better creep resistance.

[0172] By limiting the distance at which the center of the projection of the weak portion **21** in the thickness direction deviates from the center point **2213** in the range of 0-3 mm (excluding 0), the weak portion **21** has better creep resistance. At the same time, the thickness of the weak portion **21** is relatively large, so that when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, which is beneficial to reduce the risk of the weak portion **21** being damaged prematurely.

[0173] In some embodiments, the wall portion **221** includes a main body portion, and the weak portion **21** and the main body portion are integrally formed.

[0174] Integrally formed means that the weak portion **21** and the main body portion are an integral structure when provided. For example, the weak portion **21** may be formed on the wall portion **221**

by stamping or cold heading or the like.

[0175] Integrally forming the weak portion **21** with the main body portion can improve the stiffness of the weak portion **21**, so that the weak portion **21** has a stronger ability to resist creep and impact of an electrolytic solution, which is beneficial to prolong the service life of the weak portion **21** and reduce the risk of liquid leakage occurring to the weak portion **21**.

[0176] In some embodiments, the hardness of the weak portion **21** is greater than that of the main body portion.

[0177] The weak portion **21** can be formed by stamping, so that the weak portion **21** will undergo cold work hardening (the grain arrangement changes, resulting in lattice distortion, reducing the metal plasticity, and increasing the material hardness). The hardness of the weak portion **21** will be greater than the hardness of the main body portion, so that the weak portion has an enhanced ability to resist external impact, and is not likely to be damaged by external impact. In addition, the weak portion **21** has an enhanced ability to resist fatigue and is less likely to fail due to fatigue.

[0178] In some other embodiments, the wall portion **221** includes a main body portion and a pressure relief mechanism, and the pressure relief mechanism is arranged separately from the main body portion and connected thereto. The weak portion **21** is arranged on the pressure relief mechanism, and an outer surface of the main body portion is the outer surface of the wall portion **221**.

[0179] “The pressure relief mechanism is arranged separately from the main body portion and connected thereto” means that the pressure relief mechanism and the main body portion are provided separately and are connected together in the end. For example, the pressure relief mechanism may be welded to the main body portion. In this case, the pressure relief mechanism may protrude from the outer surface of the main body portion. In this case, the outer surface of the main body portion serves as the outer surface of the wall portion **221**.

[0180] The pressure relief mechanism is arranged separately from the main body portion and installed on the main body portion, so as to facilitate processing and manufacturing.

[0181] Referring to FIG. 13, FIG. 15, and FIG. 16, FIG. 15 is a side view of a battery cell **20** according to yet still some other embodiments of the present application. FIG. 16 is a schematic structural view of a battery cell **20** according to yet still some further other embodiments of the present application. In some embodiments, the shell **22** includes a case **222** and an end cover **223**, the case **222** has an opening, and the end cover **223** closes the opening and is fixed to the case **222**. The wall portion **221** is the end cover **223** or a wall of the case **222**.

[0182] Referring to FIG. 13, in the embodiments shown in FIG. 13, the end cover **223** is the wall portion **221**.

[0183] Referring to FIG. 15, in the embodiments shown in FIG. 15, the case **222** includes an integrally formed side wall **2221** and bottom wall **2222**. The side wall **2221** is arranged on all sides of the bottom wall **2222** in a defining manner, and in a thickness direction of the bottom wall **2222**, one end of the side wall **2221** is connected to the bottom wall **2222**, and the other end thereof forms an opening in a defining manner. The end cover **223** closes the opening. The side wall **2221** is the wall portion **221**.

[0184] When the case **222** is provided, the side wall **2221** and the bottom wall **2222** are of an integral structure. The side wall **2221** and the bottom wall **2222** may be integrally formed by stamping or casting. One end of the side wall **2221** is connected to the bottom wall **2222**, and the other end of the side wall **2221** is connected to the end cover **223**. In the thickness direction of the end cover **223**, the bottom wall **2222** and the end cover **223** are arranged opposite to each other.

[0185] Referring to FIG. 16, in the embodiments shown in FIG. 16, the shell **22** includes a case **222** and an end cover **223**. The case **222** includes an integrally formed side wall **2221** and bottom wall **2222**. The side wall **2221** is arranged on all sides of the bottom wall **2222** in a defining manner, and in a thickness direction of the bottom wall **2222**, one end of the side wall **2221** is connected to the bottom wall **2222**, and the other end thereof forms an opening in a defining manner. The end cover

223 closes the opening. The bottom wall **2222** is the wall portion **221**.

[0186] The thickness of the bottom wall **2222** is usually larger, but the bottom wall is also more susceptible to external impact. Therefore, by arranging the weak portion **21** in the first region **2211** of the bottom wall **2222** to increase the thickness of the weak portion **21**, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, which is beneficial to reducing the risk of the weak portion **21** being opened prematurely.

[0187] The wall portion **221** may be the end cover **223**, or the bottom wall **2222** of the case **222**, or the side wall **2221** of the case **222**, and the weak portion **21** is arranged in the first region **2211** of the end cover **223**, the bottom wall **2222** or the side wall **2221**. Under the condition of the same fracture initiation pressure, the thickness of the weak portion **21** arranged in the first region **2211** can be larger. In this way, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, which is helpful to reduce the risk of the weak portion **21** being damaged prematurely.

[0188] An embodiment of the present application further provides a battery **100**. The battery **100** includes the above battery cell **20**.

[0189] An embodiment of the present application further provides an electrical device. The electrical device includes the above battery **100**. The battery **100** is configured to provide electrical energy for the electrical device.

[0190] According to some embodiments of the present application, reference can be made to FIG. 3 to FIG. 16.

[0191] Embodiments of the present application provides a battery cell **20**, where the battery cell **20** includes a shell **22**, the shell **22** has a wall portion **221**, the wall portion **221** includes a weak portion **21**, the weak portion **21** is configured to be destroyed when the battery cell **20** releases internal pressure, and an outer surface of the wall portion **221** has a center point **2213**. A maximum distance between a projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** is a , the minimum radial dimension of the outer surface of the wall portion **221** is A , a and A satisfy: $a \leq 0.4A$, and the radial direction refers to a direction passing through the center point **2213**. By setting $a \leq 0.4A$, namely, the maximum distance between the projection of the weak portion **21** on the outer surface of the wall portion **221** and the center point **2213** of the outer surface of the wall portion **221** is less than or equal to 0.4 times the minimum radial dimension of the outer surface of the wall portion **221**, the weak portion **21** is arranged in the first region **2211**, with a relatively small stiffness, of the wall portion **221**. The first region **2211** is a circular region with the center point **2213** of the outer surface of the wall portion **221** as a circle center and 0.4 times the minimum radial dimension of the outer surface of the wall portion **221** as a radius. The first region **2211** is close to the center of the wall portion **221**, and the first region **2211** is a region, with a relatively small stiffness, of the wall portion **221** and thus has a relatively weak ability to resist deformation. When the battery cell **20** releases the internal pressure, the first region **2211** undergoes a large deformation under the action of gas, and therefore the weak portion **21** arranged in the first region **2211** is easily deformed and damaged. Therefore, under the condition of the same fracture initiation pressure, the thickness of the weak portion **21** arranged in the first region **2211** can be larger. In this way, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, which is beneficial to reduce the risk of the weak portion **21** being damaged prematurely. Furthermore, the thicker the weak portion **21** is, the easier it is to manufacture, and the lower the precision requirement for a manufacturing device.

[0192] The wall portion **221** is provided with a score groove **2212**, and the weak region **21** is formed at a bottom of the score groove **2212**. The weak portion **21** is formed by forming the score groove **2212** in the wall portion **221**, with operation simple and convenient and costs low.

[0193] The projection of the weak portion **21** on the outer surface of the wall portion **221** passes through the center point **2213**. When the battery cell **20** releases the internal pressure, the center point **2213** is a position, with the largest deformation, of the wall portion **221** and is most likely to

be damaged. By making the projection of the weak portion **21** on the outer surface of the wall portion **221** pass through the center point **2213**, the thickness of the weak portion **21** at the center point **2213** can be larger. In this way, when the battery cell **20** is normally used, the weak portion **21** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely.

[0194] The weak portion **21** has a weak point **2111** arranged corresponding to the center point **2213**, the thickness of the weak portion **21** at the weak point **2111** is a first thickness, and the maximum thickness of the weak portion **21** at other positions than the weak point **2111** is a second thickness. The first thickness is greater than or equal to the second thickness. The first thickness is greater than the second thickness, namely, the thickness of the weak portion **21** at the weak point **2111** is greater than the maximum thickness of the weak portion **21** at other positions than the weak point **2111**. The thickness of the weak portion **21** at the weak point **2111** is greater than the thickness of the weak portion **21** at other positions, and the weak point **2111** has a stronger ability to resist external impact, so as to reduce the risk of the weak portion **21** being damaged prematurely. The first thickness is equal to the second thickness, namely, the thickness of the weak portion **21** at the weak point **2111** is equal to the maximum thickness of the weak portion **21** at other positions than the weak point **2111**. In this way, the thickness of the weak portion **21** is relatively more uniform, thereby facilitating processing and manufacturing.

[0195] The above descriptions are merely preferred embodiments of the present application and are not intended to limit the present application. For those skilled in the art, the present application may have various modifications and changes. Any modification, equivalent replacement, improvement, etc. made within the spirit and principle of the present application shall fall within the scope of protection of the present application.

Claims

1. A battery cell, comprising: a shell having a wall portion, wherein the wall portion comprises a weak portion, the weak portion is configured to be destroyed when the battery cell releases internal pressure, and an outer surface of the wall portion has a center point; and a maximum distance between a projection of the weak portion on the outer surface of the wall portion and the center point is a , the minimum radial dimension of the outer surface of the wall portion is A , a and A satisfy: $a \leq 0.4A$, and the radial direction refers to a direction passing through the center point.
2. The battery cell according to claim 1, wherein the wall portion is provided with a score groove, and the weak portion is formed at a bottom of the score groove.
3. The battery cell according to claim 2, wherein the projection of the weak portion on the outer surface of the wall portion passes through the center point.
4. The battery cell according to claim 3, wherein the weak portion has a weak point arranged corresponding to the center point, the thickness of the weak portion at the weak point is a first thickness, the maximum thickness of the weak portion at other positions than the weak point is a second thickness, and the first thickness is greater than or equal to the second thickness.
5. The battery cell according to claim 4, wherein the score groove comprises at least one groove section, the weak portion comprises at least one weak section, and the at least one groove section and the at least one weak section are arranged in a one-to-one correspondence.
6. The battery cell according to claim 5, wherein the at least one weak section comprises at least one first weak section, the weak point is arranged on the first weak section, the thickness of the first weak section at the weak point is the first thickness, the thickness of the first weak section at other positions than the weak point is a third thickness, and the first thickness is greater than or equal to the third thickness.
7. The battery cell according to claim 6, wherein the at least one weak section further comprises a second weak section and a third weak section, the second weak section and the third weak section

- are arranged opposite to each other and spaced apart, the first weak section is configured to connect the second weak section and the third weak section, the thickness of the second weak section and the thickness of the third weak section are both a fourth thickness, and the fourth thickness is less than the third thickness.
- 8.** The battery cell according to claim 7, wherein one end of the first weak section is connected to a middle of the second weak section, the other end of the first weak section is connected to a middle of the third weak section, and the weak point is located at a midpoint of the first weak section.
- 9.** The battery cell according to claim 7, wherein the second weak section comprises a first end and a second end, the third weak section comprises a third end and a fourth end, the first end and the third end are arranged opposite to each other, the second end and the fourth end are arranged opposite to each other, the first weak section is configured to connect the first end and the fourth end or connect the second end and the third end, and the weak point is located at the midpoint of the first weak section.
- 10.** The battery cell according to claim 7, wherein the at least one weak section further comprises a fourth weak section and a fifth weak section, the fourth weak section and the fifth weak section are arranged opposite to each other and spaced apart, the second weak section and the fourth weak section intersect at the first end, the third weak section and the fifth weak section intersect at the second end, the first weak section is configured to connect the first end and the second end, and the weak point is located at the midpoint of the first weak section.
- 11.** The battery cell according to claim 8, wherein the at least one weak section further comprises a sixth weak section, and the first weak section and the sixth weak section intersect at the weak point.
- 12.** The battery cell according to claim 1, wherein the projection of the weak portion on the outer surface of the wall portion is a closed structure, and the closed structure surrounds the center point.
- 13.** The battery cell according to claim 1, wherein in a thickness direction of the wall portion, the center of the projection of the weak portion coincides with the center point.
- 14.** The battery cell according to claim 1, wherein in the thickness direction of the wall portion, the center of the projection of the weak portion deviates from the center point.
- 15.** The battery cell according to claim 14, wherein a distance at which the center of the projection of the weak portion in the thickness direction deviates from the center point is B, which satisfies: $0 < B \leq 5 \text{ mm}$.
- 16.** The battery cell according to claim 1, wherein the wall portion comprises a main body portion, and the weak portion and the main body portion are integrally formed.
- 17.** The battery cell according to claim 16, wherein the hardness of the weak portion is greater than that of the main body portion.
- 18.** The battery cell according to claim 1, wherein the wall portion comprises a main body portion and a pressure relief mechanism, the pressure relief mechanism is arranged separately from the main body portion and connected thereto, the weak portion is arranged on the pressure relief mechanism, and an outer surface of the main body portion is the outer surface of the wall portion.
- 19.** A battery, comprising the battery cell according to claim 1.
- 20.** An electrical device, comprising the battery according to claim 19, wherein the battery is configured to provide electrical energy for the electrical device.
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